MA3521 — Linear ODEs

Name: Answer Key

Problem 1 $(1+t^2)y' + 2ty = 2t\cos(t^2) \qquad y(0) = 2$

1a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\frac{2t}{1+t^2}$ $\qquad q(t) =$ $\frac{2t\cos(t^2)}{1+t^2}$

1b. Find an integrating factor.

$\mu(t) = \exp\left(\int \frac{2t}{1+t^2}\, dt\right) =$ $1+t^2$

1c. Multiply by $\mu(t)$ and integrate.

$(\mu y)' =$ $2t\cos(t^2)$

$\mu(t)y(t) = \int 2t\cos(t^2)\, dt + C =$ $\sin(t^2) + C$

1d. Use the initial condition to evaluate C.

$C =$ 2

$(1+t^2)y' + 2ty = 2t\cos(t^2)$

$\frac{(1+t^2)y'}{1+t^2} + \frac{2ty}{1+t^2} = \frac{2t\cos(t^2)}{1+t^2}$

$y' + \frac{2t}{1+t^2}y = \frac{2t\cos(t^2)}{1+t^2}$ — 1a

$\mu(t) = e^{\int \frac{2t}{1+t^2}}$

$u = 1+t^2$

$du = 2t\, dt \rightarrow \int \frac{2t}{1+t^2}\, dt = \int \frac{1}{u}\, du$

$= \ln|u|$

$= \ln(1+t^2)$ — plug in $u = 1+t^2$ — 1b

$\mu(t) = e^{\ln(1+t^2)}$

$= 1+t^2$

$y' + \frac{2t}{1+t^2}y = \frac{2t\cos(t^2)}{1+t^2}$ — multiply by $\mu(t)$

$(1+t^2)y' + 2ty = 2t\cos(t^2)$ — product rule

$((1+t^2)y)' = 2t\cos(t^2)$

$\int ((1+t^2)y)' = \int 2t\cos(t^2)\, dt + C$

$(1+t^2)y = \int 2t\cos(t^2)\, dt + C$

$u = t^2$

$du = 2t\, dt \rightarrow \int \cos(u)\, du$

$= \sin(u)$ — plug in $u = t^2$

$= \sin(t^2)$

$(1+t^2)y = \sin(t^2) + C$ — 1c

$y(0) = 2$ gives $y = 2$, $t = 0$

PLUG IN

$(1+0^2)2 = \sin(0^2) + C$

$(1)2 = 0 + C$

$2 = C$ — 1d

Problem 2 $(2+t^3)y' + 3t^2y = 3t^2e^{t^3} \qquad y(0) = 1$

2a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\dfrac{3t^2}{2+t^3}$ $\qquad q(t) =$ $\dfrac{3t^2e^{t^3}}{2+t^3}$

2b. Find an integrating factor.

$\mu(t) = \exp\left(\int \dfrac{3t^2}{2+3t^3}\, dt\right) =$ $2+t^3$

2c. Multiply by $\mu(t)$ and integrate.

$(\mu y)' =$ $3t^2e^{t^3}$

$\mu(t)y(t) = \int 3t^2e^{t^3}\, dt + C =$ $e^{t^3} + C$

2d. Use the initial condition to evaluate C.

$C =$ 1

$(2t+3)y' + 3t^2y = 3t^2e^{t^3}$

$\dfrac{(2t+3)y'}{(2t+3)} + \dfrac{3t^2y}{2t+3} = \dfrac{3t^2e^{t^3}}{2t+3}$

$y' + \dfrac{3t^2y}{2t+3} = \dfrac{3t^2e^{t^3}}{2t+3}$ } 2a

$M(t) = e^{\int \frac{3t^2}{2+t^3}}$

$u = 2+t^3$

$du = 3t^2 \rightarrow \int \frac{1}{u}\, du$

$= \ln|u|$

$= \ln|2+t^3|$

$M(t) = e^{\ln(2+t^3)}$

$= 2+t^3$ } 2b

$y' + \dfrac{3t^2}{2+t^3}y = \dfrac{3t^2e^{t^3}}{2+t^3}$

$(2+t^3)y' + 3t^2y = 3t^2e^{t^3}$

$((2+t^3)y)' = 3t^2e^{t^3}$

$\int ((2+t^3)y)' = \int 3t^2e^{t^3}\, dt + C$

$(2+t^3)y = \int 3t^2e^{t^3}\, dt + C$

$u = t^3$

$du = 3t^2dt \rightarrow \int e^u\, du$

$= e^u$

$= e^{t^3}$

$(2+t^3)y = e^{t^3} + C$ } 2c

$y(0) = 1$ gives $t = 0, y = 1$

$(2+0^3)1 = e^{0^3} + C$

$2 = 1 + C$

$1 = C$ } 2d